

TIANHE, China

WMO: 574940

Lat: 30.5977N

Lon: 114.0506E

Elev: 25

StdP: 101.03

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.5	-1.1	-11.9	1.4	3.8	-8.9	1.8	3.8	7.0	3.7	6.3	4.0	2.0	20	0.377

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.0	36.2	27.6	35.2	27.3	34.1	27.0	29.3	33.7	28.7	32.9	28.0	32.1	3.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.2	24.5	31.9	27.8	23.9	31.6	27.0	22.8	30.9	97.2	33.5	93.6	32.8	90.6	32.4	32.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.1	6.0	5.2	-4.6	37.9	1.8	1.3	-5.9	38.8	-7.0	39.5	-8.0	40.2	-9.3	41.1		
			-6.0	30.3	1.5	1.0	-7.0	31.0	-7.9	31.5	-8.8	32.1	-9.8	32.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.7	4.4	7.1	12.1	18.1	22.9	26.4	29.6	28.9	24.7	19.0	12.6	6.6
	DBStd	9.26	3.13	3.95	4.24	3.93	3.07	2.71	2.75	2.94	3.21	3.34	3.93	3.08
	HDD10.0	428	176	96	26	1	0	0	0	0	0	19	111	
	HDD18.3	1570	432	314	198	51	3	0	0	1	32	176	363	
	CDD10.0	3256	2	16	90	243	401	493	608	585	439	278	96	6
	CDD18.3	1357	0	1	4	43	145	243	349	327	191	52	3	0
	CDH23.3	14514	0	2	31	294	1124	2442	4595	4052	1659	300	15	0
	CDH26.7	6558	0	1	5	70	363	963	2437	2027	627	65	1	0
Wind	WSAvg	2.3	2.1	2.3	2.5	2.5	2.3	2.3	2.6	2.5	2.3	2.0	2.0	2.1
Precipitation	PrecAvg	1218	34	58	96	132	150	212	150	129	85	87	58	28
	PrecMax	1895	71	183	218	256	302	452	534	484	220	409	167	107
	PrecMin	731	1	2	17	39	36	13	29	0	3	0	1	0
	PrecStd	297	18	38	44	53	64	112	110	137	57	83	41	26
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.0	21.0	26.7	31.1	33.8	35.8	37.9	38.0	35.2	31.1	25.2	18.0
		MCWB	9.7	15.4	19.6	22.0	22.7	26.7	28.1	27.2	26.3	22.2	17.0	11.1
	2%	DB	13.4	17.8	23.2	28.7	31.9	34.1	36.7	36.2	33.2	28.2	22.8	15.2
		MCWB	8.4	12.1	16.5	20.5	22.7	26.3	27.8	27.7	25.3	20.4	15.8	9.8
	5%	DB	11.3	15.2	21.1	26.8	30.2	32.8	35.7	35.1	31.8	26.6	20.6	13.6
		MCWB	7.4	10.4	15.3	19.4	22.4	25.7	27.6	27.7	24.7	19.2	14.8	8.7
	10%	DB	9.7	13.2	19.0	24.8	28.8	31.3	34.5	33.9	30.2	24.8	18.8	11.9
		MCWB	6.3	9.3	13.8	18.6	21.8	25.0	27.4	27.2	23.8	18.5	13.8	8.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.7	16.3	20.8	24.0	26.2	29.0	30.3	30.3	28.5	24.0	19.2	12.8
		MCDB	13.9	20.0	24.9	28.7	30.5	33.2	34.6	35.0	32.5	27.5	22.2	15.8
	2%	WB	9.6	13.1	18.0	22.1	24.7	27.8	29.4	29.4	27.2	22.1	17.3	10.9
		MCDB	12.0	15.9	22.0	26.4	28.7	31.4	33.8	33.8	31.1	26.2	20.6	13.8
	5%	WB	8.1	11.2	16.2	20.8	23.9	27.0	28.7	28.7	26.3	20.8	15.9	9.7
		MCDB	10.6	14.2	19.9	24.9	27.8	30.7	33.1	32.8	30.1	24.7	19.2	12.4
	10%	WB	6.6	10.0	14.7	19.5	23.0	26.2	28.1	28.1	25.3	19.8	14.7	8.5
		MCDB	9.0	12.5	18.0	23.2	26.8	29.8	32.5	31.9	28.8	23.2	17.7	11.0
Mean Daily Temperature Range	5% DB	MDBR	6.9	7.2	8.0	8.4	8.2	7.2	7.0	7.3	7.9	8.1	8.0	7.6
		MCDBR	11.0	11.3	11.7	11.5	10.9	9.2	8.3	8.8	9.7	11.3	11.3	11.3
		MCWBR	6.5	6.3	6.1	5.3	4.6	4.0	3.4	3.6	4.1	4.7	5.4	6.3
	5% WB	MCDBR	9.5	9.4	10.3	9.8	8.9	8.1	8.0	8.2	8.5	9.3	9.2	9.8
		MCWBR	6.2	5.9	6.1	5.5	4.6	4.3	3.9	4.0	4.5	4.6	4.9	5.9
Clear-Sky Solar Irradiance	taub	0.666	0.720	0.745	0.741	0.719	0.728	0.664	0.702	0.726	0.722	0.686	0.653	
	taud	1.531	1.469	1.452	1.500	1.578	1.587	1.738	1.640	1.570	1.525	1.554	1.572	
	Ebn at Noon	542	564	595	624	644	636	676	643	601	554	523	523	
	Edh at Noon	238	276	299	294	273	270	231	252	261	257	231	219	
All-Sky Solar Radiation	RadAvg	2.02	2.30	3.35	4.03	4.35	4.38	5.04	4.68	3.91	3.06	2.46	2.04	
	RadStd	0.35	0.58	0.25	0.44	0.43	0.45	0.52	0.49	0.43	0.41	0.31	0.39	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	-0.69	N/A	+0.56	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page